



NULLCLASS INTERNSHIP REPORT

On

Real-Time Web Development Internship Projects

Working as a Web Development Intern at **NullClass**

From 18-04-2025 to 18-07-2025

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OBJECTIVE

The Goal of this Internship was to create an Interactive and Responsive Website that helps track and understand user activity, following certain business and time limits.

INTRODUCTION

During my internship, I created an interactive and responsive website using HTML, CSS, and JavaScript. The project included building useful features like video upload with time and size limits, email OTP verification, browser notifications, forgot password with random password generator, and a reward points system. I also worked on adding subscriptions with payment integration, multi-language support, and making sure the site looked good on all devices. The website was designed to help users engage better and manage their activities easily.

BACKGROUND

Web applications today need to provide smooth, interactive, and user-friendly experiences because users expect websites to work well on all devices and offer useful features. By using HTML, CSS, and JavaScript, it is possible to build responsive websites that help users perform different actions like posting questions, uploading videos, receiving notifications, and managing their profiles.

The website I worked on aimed to support students and users by offering key features such as video upload with size and time limits, email/mobile OTP verification, browser notifications, password reset with random password generation, and a reward points system. It also included subscription plans, multi-language support, and secure payment integration.

The main challenge was to ensure that all these features worked together smoothly while keeping the website fast, responsive, and easy to use. The project was designed to help users engage better with the platform and make their experience simple and secure.

LEARNING OBJECTIVES

During this internship, I aimed to achieve the following learning objectives:

- Build practical skills in creating responsive and interactive websites using HTML, CSS, and JavaScript.
- Improve my ability to use Visual Studio Code (VS Code) efficiently for writing, testing, and debugging code.
- Learn to apply media queries and CSS techniques to make websites look good on different devices.
- Develop features like video uploads with validation, browser notifications, and password reset systems.
- Implement user authentication flows, including OTP verification for email and mobile.
- Integrate basic payment gateway functionality and subscription models.
- Apply best practices in writing clean, well-structured, and maintainable front-end code.
- Strengthen problem-solving and critical thinking by building user-friendly and secure web features.

ACTIVITIES AND TASKS

During the internship, I undertook the following activities and tasks:

- Designed and developed responsive web pages using HTML, CSS, and JavaScript.
- Used VS Code for coding, debugging, and managing the project files efficiently.
- Added media queries and extra CSS to make the website work well on different devices.
- Implemented video upload feature with validations for file size, duration, and upload timing.
- Created OTP-based authentication for secure video uploads and language switching.
- Built browser notifications for user activities like answers and upvotes, with enable/disable options.
- Developed a forgot password page with random password generation using only uppercase and lowercase letters.
- Added a reward points system for user contributions, with point transfers and reductions based on actions.
- Integrated payment gateway options for subscriptions with different posting limits and time restrictions.
- Enabled multi-language support with email or mobile OTP verification during language switch.

SKILLS AND COMPETENCIES

Through this internship, I enhanced and developed the following skills and competencies:

- **Technical Skills:** HTML, CSS, JavaScript, media queries, browser Notification API, OTP authentication, and using Visual Studio Code for development and debugging.
- **Analytical Skills:** Understanding user needs, designing interactive features, and solving problems related to responsiveness and functionality.
- **Time Management:** Managing tasks efficiently to meet deadlines and handle multiple features like video upload, rewards, and subscriptions.
- **Attention to Detail:** Writing clean code, ensuring proper validation, and checking feature conditions (e.g., time limits, file sizes).
- **Critical Thinking:** Building secure and efficient solutions for password reset, point transfers, and subscription rules.
- **Communication:** Documenting code clearly and explaining features to team members and mentors.

FEEDBACK AND EVIDENCE

My internship experience was marked by valuable constructive feedback, which I actively sought and appreciated. My mentors particularly commended:

- My ability to design and implement responsive web pages using HTML, CSS, and JavaScript.
- The accuracy and reliability of the validation logic applied for features like video uploads, OTP authentication, and password reset.
- My consistent effort in clarifying requirements and sharing ideas during discussions and meetings.
- My proactive approach to applying feedback and quickly improving features based on suggestions.

The culmination of my efforts was the successful development and delivery of a fully functional and interactive web application. This deliverable effectively met all defined requirements by providing a responsive interface, secure user flows, and useful features such as video uploads, notifications, and a reward system.

CHALLENGES AND SOLUTIONS

During the internship, I faced several challenges and here is how I overcame them:

- **Challenge:** Making the website fully responsive across different devices.
Solution: Used media queries and additional CSS rules to adjust layouts and elements for various screen sizes, ensuring a smooth user experience.
- **Challenge:** Implementing secure and time-restricted video uploads.
Solution: Added validations in JavaScript to check file size, duration, and upload timing before accepting the video, along with OTP verification.
- **Challenge:** Managing browser notifications effectively.
Solution: Used the Notification API with permission checks and built options for users to enable or disable notifications from their profile.
- **Challenge:** Handling password reset without using special characters or numbers.
Solution: Created a random password generator in JavaScript that outputs passwords using only uppercase and lowercase letters.
- **Challenge:** Adding a payment gateway with time-restricted transactions.
Solution: Integrated payment options (e.g., Stripe, Razorpay) with checks to allow payments only within the allowed time frame and sent invoices via email.

OUTCOMES AND IMPACT

The successful completion of this internship led to several key achievements:

- Development of a fully functional and responsive web application using HTML, CSS, and JavaScript.
- Strengthened skills in building interactive features such as secure video uploads, notifications, and reward systems.
- Gained significant experience in managing user authentication flows and payment integrations.
- Improved ability to create time-based validations and controls for features like video uploads and payments.
- Enhanced problem-solving skills by addressing real-world challenges in web development and ensuring a smooth user experience across devices.

CONCLUSION

This internship provided me with significant growth in web development using HTML, CSS, JavaScript, and VS Code. I successfully built an interactive and responsive web application that included advanced features such as secure video uploads with time and size restrictions, OTP-based authentication, real-time notifications, reward systems, and subscription plans with payment gateway integration. These tasks allowed me to enhance my technical skills and problem-solving abilities while working on practical challenges in frontend and backend web development. The knowledge and experience gained through this project will be highly valuable for my future career as a web developer. Overall, this enriching internship offered valuable learning opportunities that strengthened my expertise in creating user-friendly and functional web applications.